

EXPERIMENT MANUAL

Basic Electrical Engineering Laboratory

For B.Tech 1st / 2ND Semesters

Under

Girijananda Choudhury University

Department of Electrical Engineering



GIRIJANANDA CHOWDHURY UNIVERSITY, ASSAM
Guwahati-781017

LABORATORY PRACTICE

SAFETY RULES

1. SAFETY is of paramount importance in the Electrical Engineering Laboratories.
2. Electricity NEVER EXECUSES careless persons. So, exercise enough care and attention in handling electrical equipment and follow safety practices in the laboratory. (Electricity is a good servant but a bad master).
3. Avoid direct contact with any voltage source and power line voltages. (Otherwise, any such contact may subject you to electrical shock)
4. Wear rubber-soled shoes. (To insulate you from earth so that even if you accidentally contact a live point, current will not flow through your body to earth and hence you will be protected from electrical shock)
5. Wear laboratory-coat and avoid loose clothing. (Loose clothing may get caught on an equipment/instrument and this may lead to an accident particularly if the equipment happens to be a rotating machine)
6. Girl students should have their hair tucked under their coat or have it in a knot.
7. Do not wear any metallic rings, bangles, bracelets, wristwatches and neck chains. (When you move your hand/body, such conducting items may create a short circuit or may touch a live point and thereby subject you to electrical shock)
8. Be certain that your hands are dry and that you are not standing on wet floor. (Wet parts of the body reduce the contact resistance thereby increasing the severity of the shock)
9. Ensure that the power is OFF before you start connecting up the circuit.(Otherwise you will be touching the live parts in the circuit)
10. Get your circuit diagram approved by the staff member and connect up the circuit strictly as per the approved circuit diagram.
11. Check power chords for any sign of damage and be certain that the chords use safety plugs and do not defeat the safety feature of these plugs by using ungrounded plugs.
12. When using connection leads, check for any insulation damage in the leads and avoid such defective leads.
13. Do not defeat any safety devices such as fuse or circuit breaker by shorting across it. Safety devices protect YOU and your equipment.

14. Switch on the power to your circuit and equipment only after getting them checked up and approved by the staff member.
15. Do not make any change in the connection without the approval of the staff member.
16. In case you notice any abnormal condition in your circuit (like insulation heating up, resistor heating up etc), switch off the power to your circuit immediately and inform the staff member.
17. Keep hot soldering iron in the holder when not in use.
18. After completing the experiment show your readings to the staff member and switch off the power to your circuit after getting approval from the staff member.
19. Remove all the connections of the equipments and devices after performing the experiments.
20. Some students have been found to damage meters by mishandling in the following ways:
 - (a) Keeping unnecessary material like books, lab records, unused meters etc. causing meters to fall down the table.
 - (b) Putting pressure on the meter (specially glass) while making connections or while talking or listening somebody.

STUDENTS ARE STRICTLY WARNED THAT FULL COST OF THE METER WILL BE RECOVERED FROM THE INDIVIDUAL WHO HAS DAMAGED IT IN SUCH A MANNER.

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Guidelines for laboratory notebook

Trouble shooting hints

BEE LABORATORY EXPERIMENT

Serial No	Experiments
1.	Basic safety precaution, introduction and use of measuring instruments.
2.	Calibration of Milliammeter as a voltmeter.
3.	Calibration of Millivoltmeter as an Ammeter.
4.	Verification of Thevenin's Theorem.
5.	Verification of Maximum Power Transfer Theorem.
6.	Study of R-L-C Series Circuit.
7.	Forward Characteristics of Semiconductor Diode.
8.	Measurement of Power in a Single Phase AC circuit using Wattmeter.
9.	Demonstration of layout of house wiring.
10	Demonstration of measurement of insulation resistance.

DATA-SHEET

EXPERIMENT NO: 1

TITLE: BASIC SAFTY PRECAUTIONS, INTRODUCTION AND USE OF MEASURING INSTRUMENTS.

OBJECTIVES:To study the

- (i) Basic safety precaution and
- (ii) Introduction and use of measuring instruments.

(i) BASIC SAFTY RULES :-

1. Avoid direct contact with any voltage source and power line voltages. (Otherwise, any such contact may subject you to electrical shock)
2. Wear rubber-soled shoes. (To insulate you from earth so that even if you accidentally contact a live point, current will not flow through your body to earth and hence you will be protected from electrical shock)
3. Wear laboratory-coat and avoid loose clothing. (Loose clothing may get caught on an equipment/instrument and this may lead to an accident particularly if the equipment happens to be a rotating machine)
4. Girl students should have their hair tucked under their coat or have it in a knot.
5. Do not wear any metallic rings, bangles, bracelets, wristwatches and neck chains. (When you move your hand/body, such conducting items may create a short circuit or may touch a live point and thereby subject you to electrical shock)
6. Be certain that your hands are dry and that you are not standing on wet floor. (Wet parts of the body reduce the contact resistance thereby increasing the severity of the shock)
7. Ensure that the power is OFF before you start connecting up the circuit.(Otherwise you will be touching the live parts in the circuit)

(ii) INTRODUCTION AND USE OF MEASURING INSTRUMENTS:-

The instruments used to measure any quantity are known as measuring instruments. If the instruments can measure the basic electrical quantities, such as voltage and current are known as basic measuring instruments.

TYPES OF BASIC MEASURING INSTRUMENTS:-

Basic measuring instruments into the following two types---

(A) VOLTMETER

(B) AMMETER

Let us discuss about these two basic measuring instruments briefly.

(A) VOLTMETER :-voltage measuring instrument is a measuring instrument which measures the voltage across any two points of an electric circuit. The units of voltage are volt and the measuring instrument is meter. Hence, the word “voltmeter” is obtained by combining the two words “volt” and “meter”.

We can classify the voltmeters into the following two types based on the type of voltage that it can measure.

- DC Voltmeters (PMMC)
- AC Voltmeters (MI)

DC VOLTMETER:-

DC voltmeter measures the DC voltage across any two points of an electric circuit. A practical DC voltmeter is shown in below figure.



The DC voltmeter shown in the figure is a (0–10)V DC voltmeter. Hence, it can be used to measure the DC voltages from zero volts to 10 volts

AC VOLTMETER:-

AC voltmeter measures the AC voltage across any two points of an electric circuit. A practical AC voltmeter is shown in below figure.



The AC voltmeter shown in above figure is a (0–250)V AC voltmeter. Hence, it can be used to measure the AC voltages from zero volts to 250 volts

(B)AMMETER:- Ammeter is a measuring instrument which measures the current flowing through any two points of an electric circuit. The unit of current is ampere and the measuring instrument is meter. The word “ammeter” is obtained by combining “am” of ampere with “meter”.

We can classify the ammeters into the following two types based on the type of current that it can measure.

- DC Ammeter (PMMC)
- AC Ammeter (MI)

DC AMMETER

As the name suggests, DC ammeter measures the DC current that flows through any two points of an electric circuit. A practical DC ammeter is shown in figure.



The DC ammeter shown in above figure is a (0–50)A(0–50)A **DC ammeter**. Hence, it can be used to measure the DC currents from zero Amperes to 50 Amperes

AC AMMETER

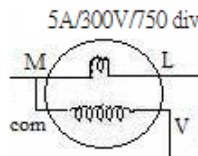
AC ammeter measures the AC current that flows through any two points of an electric circuit. A practical AC ammeter is shown in below figure.



The AC ammeter shown in above figure is a (0–10)A AC ammeter. Hence, it can be used to measure the AC currents from zero Amperes to 10 Amperes.

Electrodynamometer Wattmeter :-

The instrument whose working depends on the reaction between the magnetic field of moving and fixed coils is known as the Electro dynamo-meter Wattmeter. It uses for measuring the power of both the AC and DC circuits.



Digital Multimeter:-

A digital multimeter (DMM) is a test tool used to measure two or more electrical values principally voltage (volts), current (amps) and resistance (ohms). It is a standard diagnostic tool for technicians in the electrical/electronic industries.

Digital multimeters combine the testing capabilities of single-task metersthe voltmeter (for measuring volts), ammeter (amps) and ohmmeter (ohms).

The face of a digital multimeter typically includes four components:-

- Display: Where measurement readouts can be viewed.
- Buttons: For selecting various functions; the options vary by model.
- Dial (or rotary switch): For selecting primary measurement values (volts, amps, ohms).
- Input jacks: Where test leads are inserted.



EXPERIMENT NO:2

TITLE: CALIBRATION OF MILLIAMMETER AS A VOLTMETER.

OBJECTIVES: To calibrate a milliammeter as a voltmeter with the help of a substandard voltmeter and draw the calibration curve and Correction curve for the given milliammeter.

CIRCUIT DIAGRAM:

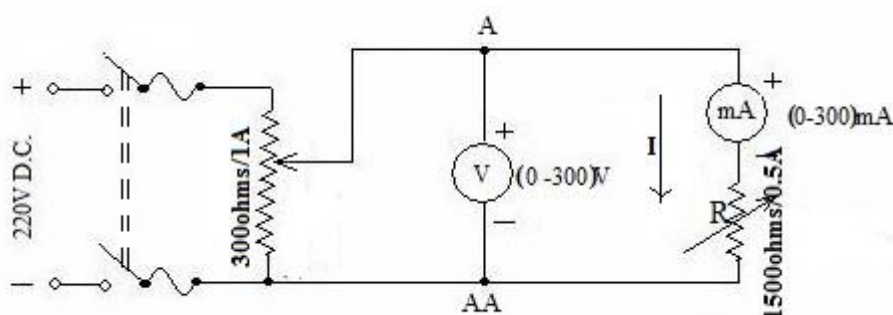


Fig.1

LIST OF APPARATUS:

Serial No.	Apparatus	Maker	Type	Range	Quantity
1	Ammeter		PMMC	0-300 mA	1
2	Voltmeter		PMMC	0-300 V	1
3	Rheostat		Wire wound	300 Ohms/1Amp	1
4	Rheostat		Wire wound	1500 Ohm/ 0.5 Amp	1
5	Power Supply		DC	220 V	1

THEORY:

Let, R be the external variable resistor connected in series with the milliammeter shown in Fig.1. The internal resistance of the milliammeter is (say) R_a . If E be the voltage applied across the nodes A & AA and I be the current through the combination of the milliammeter and R then we can write:

$$\frac{E}{I} = (R + R_a)$$

If V volts is the voltmeter reading corresponding to the full scale reading of the milliammeter (I_n milliamps) then we can define a constant

$$K = V/I_n$$

Hence for any milliammeter reading I:

The equivalent voltage $V_{eq} = I.K (= I.V/I_n)$

Here the milliammeter is the ***meter under test*** while the voltmeter is the ***substandard meter***. That is we are comparing the readings of the milliammeter (used here as a voltmeter) with those of the voltmeter with the assumption that the voltmeter gives the more accurate readings. By comparing the readings of the two instruments we can define for the milliammeter:

$$\text{Error} = (V_{eq} - V) \text{ Volt}$$

Where , V is the substandard voltmeter reading.

Further, the correction required in the milliammeter readings, if any, is:

$$\text{Correction} = V - V_{eq} = \boxed{- \text{Error Volts}}$$

PROCEDURE:

1. Connect the circuit as shown in the circuit diagram.
2. Note the zero setting of the instruments
3. Adjust the potential divider to obtain zero output voltage and put the external resistance R at maximum value (so as to protect the milliammeter). Switch on the power supply.
4. Increase the output voltage slowly (with the help of the potential divider) upto 200V as read by the voltmeter.
5. Adjust the external resistance R to obtain the full scale deflection of the milliammeter. Make sure that the voltmeter reading is at 200Volts – fine tune the potential divider if this becomes necessary. Read and record the reading of the ammeter.
6. Decrease the output Voltage (with the help of the potential divider) in suitable steps down to zero Volts of the voltmeter. Read the voltmeter and the ammeter. Note and tabulate the instrument readings at every step. Take at least 10 sets of readings.

EXPERIMENTAL OBSERVATION AND RESULTS: $K = V/I_n$

Serial No.	Substandard voltmeter readings (V)	Milliammeter readings, I (mA)	Equivalent Voltage $V_{eq} = KI$	Correction $V - V_{eq}$	Remark

GRAPH:

Draw the **Calibration curve** and **Correction curve** for the given milliammeter in a single graph paper. (Calibration curve: I in X-axis & V in Y-axis. Correction curve: I in X-axis & correction in Y-axis.) The curve shall not be a smooth one but a “point to point” type.

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORT:

1. What do you mean by calibration? Why is it required?
2. Define equivalent voltage. Show sample calculation for V_{eq} .
3. What is the difference between an ammeter & a voltmeter? Show how a galvanometer can be used as an ammeter & as a voltmeter.
4. A moving coil galvanometer of resistance $R \Omega$ gives full scale deflection of current I amp. Find the value of resistance to be connected in series with the galvanometer to obtain full scale deflection of V volts when use as a voltmeter.

EXPERIMENT NO: 3

TITLE: CALIBRATION OF A MILLIVOLTMETER AS AN AMMETER.

OBJECTIVES: To calibrate a millivoltmeter as an ammeter with the help of a substandard voltmeter and draw the calibration curve and Correction curve for the given milliammeter.

CIRCUIT DIAGRAM:

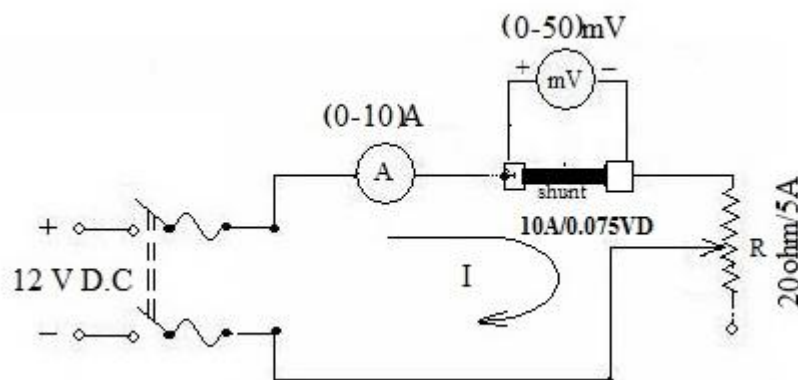


Fig.2

LIST OF APPARATUS:

Serial No.	Apparatus	Maker	Type	Range	Quantity
1	Ammeter		PMMC	0-10 A	1
2	Voltmeter		PMMC	0-50 mV	1
3	Shunt			10A/0.075VD	1
4	Rheostat		Wire wound	20 Ohm/5 Amp	1
5	Power Supply		DC	12 V	1

THEORY:

Referring to the fig2, let R_{sh} & R_m be the resistance of the shunt & the millivoltmeter respectively. Therefore R_{sh} and R_m form a parallel combination. If V be the voltage drop across the shunt and millivoltmeter combination due the circuit current I then it is easy to see that:

$$V = \left[\frac{R_{sh} R_m}{R_{sh} + R_m} \right] \times I$$

Note that the quantity with the parentheses

is simply the parallel equivalent resistance of R_m and R_{sh} which are constant numbers.

Therefore we can write:

$$V = K \cdot I \quad \text{-----} \quad (1)$$

To evaluate K, we proceed as follows:

Adjust the variable resistor R so that the millivoltmeter reads full scale. Note the corresponding ammeter reading. Then using equation (1) we can find the value of K.

Hence: $I_{eq} = E/K$ for any voltmeter reading E (say)

Further if there is a difference between the calculated and observed values of I then we define

$$\begin{aligned} \text{Error} &= I_{eq} - I \text{ (as recorded by the ammeter)} \quad \text{and} \\ \text{Correction} &= - \text{Error.} \end{aligned}$$

PROCEDURE:

1. Connect the circuit as shown in figure 1.
2. Note the zero setting of the instruments.
3. Adjust the loading rheostat R at its maximum value. Turn on power supply.
4. Increase the circuit current by slowly decreasing the load resistance until the millivoltmeter gives full-scale deflection. Note the ammeter and voltmeter readings as the first observation
5. Decrease the circuit current (with the help of the loading rheostat) in suitable steps down to zero Ampere of SM. Read and tabulate the ammeter and the voltmeter readings. Take at least 10 sets of readings.

OBSERVATION AND RESULTS:

Serial No.	Substandard ammeter readings (I) Amps	Millivoltmeter readings (V) milliVolts	$K = V/I$; V and I come from the 1 st Set of readings	$I_{eq} = V/K$	Correction $I - I_{eq}$	Remarks
1.		(full scale)			0	
2.						
3.						
4.						

GRAPH:

Draw the **Calibration curve** and **Correction curve** for the given millivoltmeter in a single graph paper. The calibration graph is to be drawn between the ammeter readings on the ordinate and the voltmeter readings on the abscissa. The curve shall not be a smooth one but a "point to

point” type. The correction graph is to be drawn between the voltmeter reading on the abscissa and the “correction” on the ordinate. Again this curve shall also be a “point to point” type.

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORT:

1. What do you mean by calibration?
2. Define equivalent ampere. Show a sample calculation for I_{eq} .
3. What is a shunt resistor and how do you specify the rating of a shunt?
4. Why does a shunt resistor have two pairs of terminals?
5. Derive an expression for the multiplying factor of a shunt resistor.

EXPERIMENT NO: 4

TITLE: VERIFICATION OF THEVENIN'S THEOREM.

OBJECTIVES:

1. To calculate Thevenin's equivalent of a given network
2. To verify Thevenin's Theorem

CIRCUIT DIAGRAM:

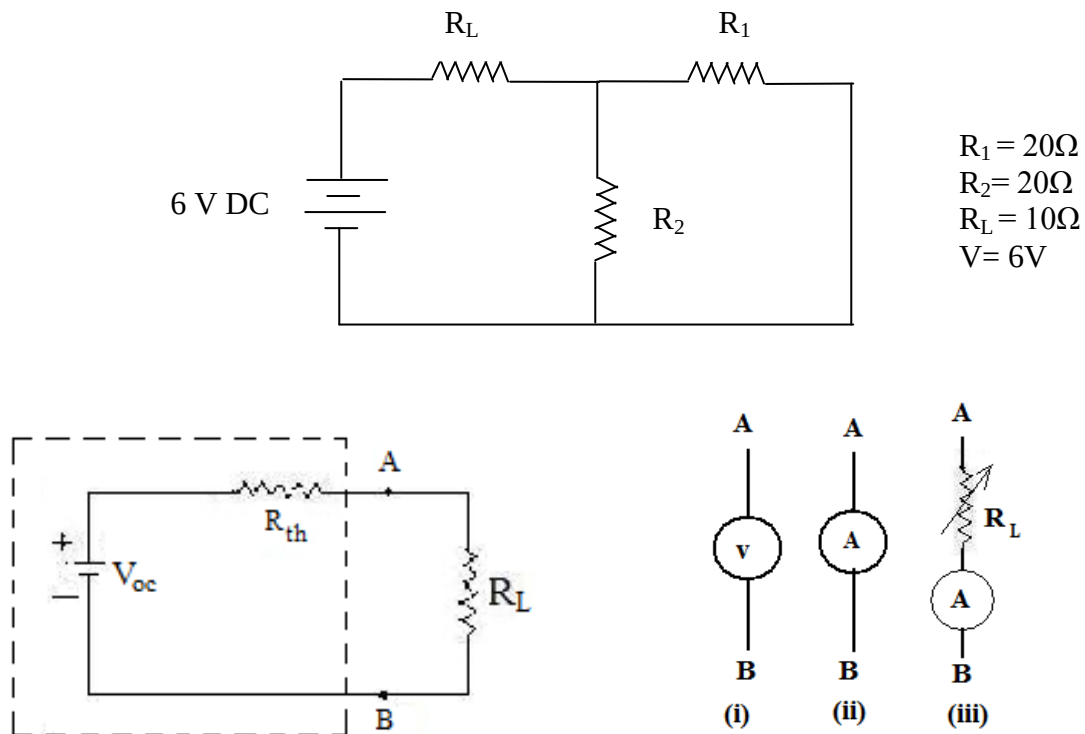


Fig. 5

LIST OF APPARATUS:

Serial No	Apparatus	Maker	Type	Range	Quantity

THEORY:

Refer textbook.

PROCEDURE:

1. Remove R_L from the given circuit.
2. Measure the open circuit voltage (V_{oc}) between A & B.
3. Measure the short circuit by connecting an ammeter A & B.
4. Measure the Thevenin's equivalent resistance R_{th} .
5. Construct Thevenin's equivalent circuit.
6. Calculate the current through R_L to be connected between A and B using the Thevenin's circuit as $I_L = V_{oc} / (R_{th} + R_L)$
7. Compare the calculated current (step 5) with the actual current by connecting an ammeter in series with the load R_L between A and B
8. Compare the analytically derived equivalent circuit with that obtained in step 5.

CALCULATIONS:

Calculate Thevenin's equivalent resistance R_{th} , V_{th} , I_{SC} and I_L manually.

OBSERVATIONS& RESULTS:

Quantity	Measured Value	Calculated value
V_{oc}		
I_{SC}		
R_{th}		
I_L		

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORT:

1. State and explain Thevenin's theorem.
2. Draw the Norton's equivalent circuit of the circuit shown in the circuit diagram.

EXPERIMENT NO: 5

TITLE: VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM.

OBJECTIVES:

1. To measure the power absorbed by the load.
2. To verify that the power absorbed by the load is maximum only when the load resistance is equal to the source resistance.

CIRCUIT DIAGRAM:

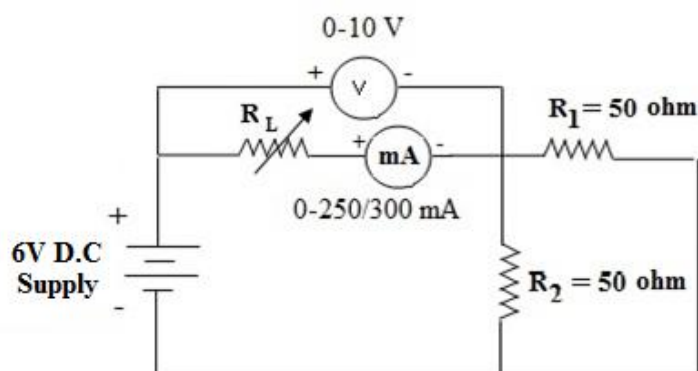


Fig. 6

LIST OF APPARATUS:

Serial No	Apparatus	Maker	Type	Range	Quantity

THEORY:

The maximum power transfer theorem states that a load will receive maximum power from a linear bilateral dc network when its total resistance value is exactly equal to the Thevenin's equivalent resistance of the network as seen by the load. Maximum power will be transferred to the load when $R_L = R_{th}$. Vary R_L from 50Ω to 5Ω

PROCEDURE:

1. Set R_L to maximum. Read Voltmeter reading (V) and Ammeter reading (I).
2. Gradually reduce R_L and at each step note V and I.
3. Stop when the current is about 100 mA.

OBSERVATIONS:

Sl No.	V (Volts)	I (mA)	R_L (ohm)	Power $P = I^2 R_L$ (watt)

GRAPH:

Plot P vs R_L and note R_L for the maximum power ($P_{\max}$). Compare this value of R_L with R_{th}

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORT:

1. State and prove “Maximum Power transfer theorem”.
2. Calculate the value of maximum power that will be absorbed by R_L .
3. Give the statement of maximum power transfer theorem in case of AC circuit.

EXPERIMENT NO: 6

TITLE: STUDY OF R-L-C SERIES CIRCUIT.

OBJECTIVES: To determine the circuit parameters, power factors and to draw the phasor diagram of a series R-L-C circuit.

CIRCUIT DIAGRAM:

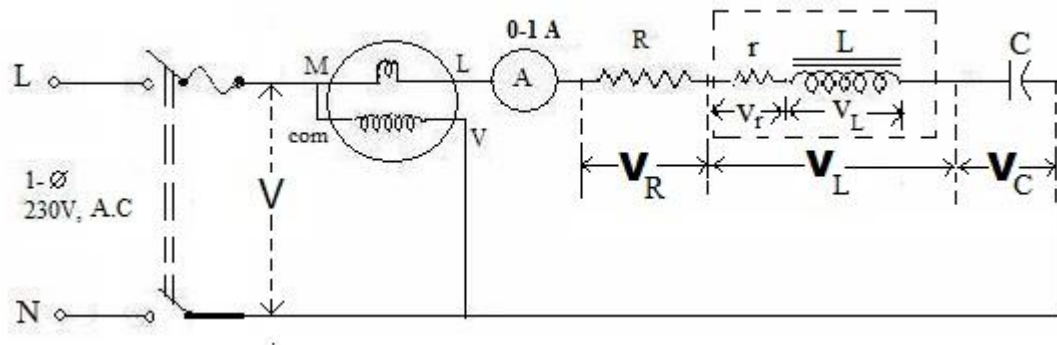
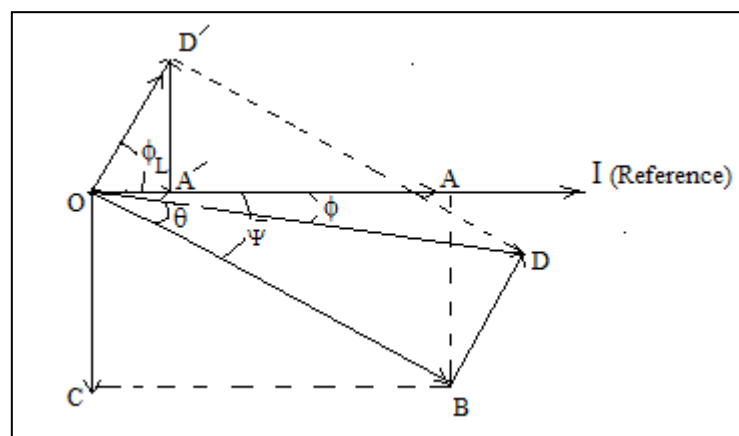


Fig.8

LIST OF APPARATUS:

Serial No	Apparatus	Maker	Type	Range	Quantity

THEORY:



V_R = Voltage drop across the resistor R
 V_C = Voltage drop across the capacitor C
 V_L = Voltage drop across the choke coil

$$OA = V_R$$

$$OC = V_C$$

$$OD = V = \text{supply voltage}$$

$$OB = V_1 = \sqrt{(V_R^2 + V_C^2)} \quad V_r = \text{In phase component of } V_L$$

$$A'D' = V_L V_L = \text{Quadrature component of } V_L$$

$$BD = OD' = V_L \quad V_1 = \text{Resultant voltage of } V_R \& V_C$$

$$OA' = V_r \quad I = \text{Line current} \& V = \text{Supply voltage}$$

ϕ = Phase angle between Current I & supply Voltage V

θ = Angle between V_1 & V.

$$\psi = \text{Angle between } V_1 \& V_R = \tan^{-1}(V_C / V_R)$$

Applying cosine law to the ΔOBD

$$\cos\theta = \frac{[OD^2 + OB^2 - BD^2]}{2 \cdot OD \cdot OB} = \frac{[V^2 + V_1^2 - V_L^2]}{2 \cdot V \cdot V_1}, \quad \theta = \cos^{-1} \frac{[V^2 + V_1^2 - V_L^2]}{2 \cdot V \cdot V_1}$$

Thus the power factor of the whole circuit, $\cos\phi = \cos(\psi - \theta)$

The in phase component of the voltage drop across the choke coil, $V_r = I \cdot r = V \cos\phi - V_R$ volt

voltage across the reactance of the choke coil, $V_L = (V_L^2 - V_r^2)^{1/2}$

Thus, inductive reactance of the choke coil, $X_L = V_L / I$.

$$\text{But, } X_L = 2\pi f L.$$

$$\text{Thus, } L = X_L / 2\pi f$$

Power absorbed by the resistance of the choke coil = $I^2 r$

$$\text{Resistance } R = V_R / I \text{ ohm}$$

$$\text{Capacitive reactance } X_C = V_C / I,$$

$$X_C = 1 / 2\pi f C \text{ and } C = 1 / 2\pi f X_C$$

$$\text{Total impedance} = \sqrt{[(R + r)^2 + (X_L - X_C)^2]}$$

$$\text{Power factor of the coil} = \cos\phi_L = r / Z_{\text{COIL}}$$

$$\text{Impedance of the coil} = Z_{\text{COIL}} = \sqrt{(r^2 + X_L^2)}$$

PROCEDURE:

1. Discharge the capacitor by shorting the terminals of the capacitor.
2. Connect the circuit as shown in the circuit diagram .
3. Take a volt meter (ac) and measure V, V_R , V_L , and V_C
4. Note the ammeter and wattmeter reading.
5. discharge the capacitor

EXPERIMENTAL OBSERVATION & RESULTS:

Voltmeter Reading (Volt)				Ammeter Reading	Wattmeter Reading	
V	V _R	V _L	V _C	Ampere	Division	Watt

Circuit parameter	Analytical	Graphical
R		
L		
C		
r		
$\cos\phi$		
$\cos\phi_L$		

CONSTRUCTION OF PHASOR DIAGRAM OF SERIES AC CIRCUIT

1. Draw a reference line OO' that represents current I of the circuit
2. Represent $OA = V_R$ (in scale) on the reference line i.e. in same phase with I
3. Draw $OC = V_C$ (in scale) perpendicular to I
4. $OB = V_1$ represents the resultant of OC and OA i.e. resultant of V_C and V_R
5. From point B draw an arc equal to V_L (in scale)
6. Draw another arc from point O equal to V (in scale) which cuts the previous arc at point D . Now $BD = V_L$ & $OD = V$
7. Draw OD' parallel and equal to BD draw a perpendicular from D' on the reference line so that $D'A' = V_L$ & $OA' = V_r$

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORT:

1. Refer theory and calculate the circuit parameters
2. Draw the phasor diagram and verify the result
3. Write down the expression for the impedance of the choke coil and circuit as a whole in complex form.
4. What is the energy stored in a capacitor.

EXPERIMENT NO: 7

TITLE: FORWARD CHARACTERISTICS OF SEMICONDUCTOR DIODE.(TYPE IN4007)

OBJECTIVES:

To plot the forward characteristics of a diode and to calculate the static and the dynamic resistance of the diode at a particular point

CIRCUIT DIAGRAM:

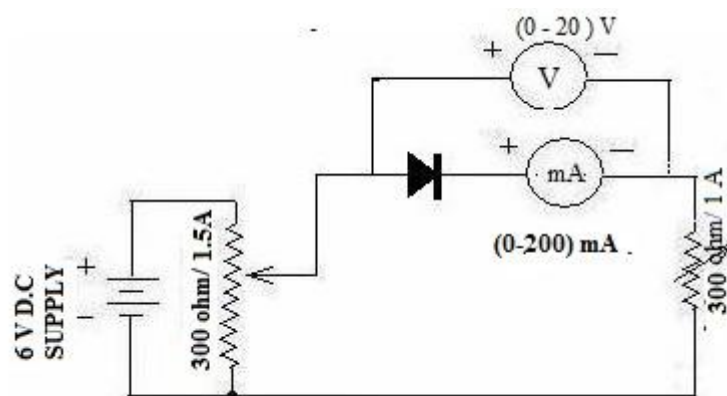


Fig. 10

LIST OF APPARATUS:

Serial No	Apparatus	Maker	Type	Range	Quantity

THEORY:

Referring to the $V-I$ characteristics curve V_Z denote the reverse biased voltage at which avalanche or Zener breakdown occurs. On the forward biased region V_o denote the voltage at which diode current begins to rise sharply. This is called Cut-in voltage or Knee voltage. V_o for Ge diode is 0.3V & for Si diode 0.7 V.

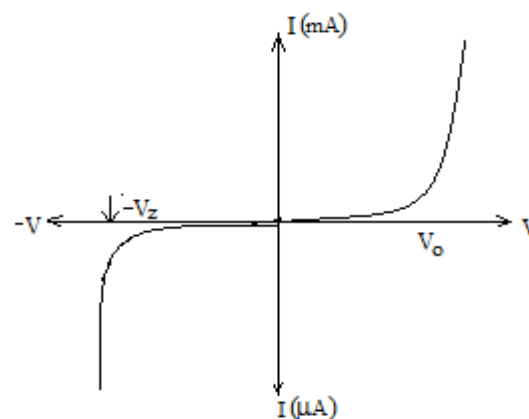


Fig. 10(a) V-I Characteristics of Diode

Forward static resistance: This can be determined from forward static characteristics, this is the ratio of voltage V_1 at a given operating point P_1 to the corresponding current I_1

Static resistance $R_S = V_1/I_1$

Dynamic resistance: It is the ratio of small change in voltage to a small change in current about the operating point

$$R_d = \Delta V / \Delta I$$

The value of R_S is of about 10 ohms & R_d is in the order of mega ohms.

PROCEDURE:

1. Study the manner in which the semiconductor device are numbered & make a note for it.
2. Connect the circuit as shown in the diagram, switch on the power supply adjusting the potential divider at zero output voltage.
3. Note the milliammeter & voltmeter reading by increasing voltage slowly with the help of potential divider in steps.
4. Draw the graph between voltage & current.
5. At a suitable operating point calculate the static & dynamic resistance of the diode.

OBSERVATION & RESULT :

SL No.	Voltage	Current

GRAPH:

Plot the V-I curve and measure the $V_{cut\,in}$ voltage. Also determine static & dynamic resistance from the graph.

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORTS:

1. Calculate the static & dynamic resistance, note that the ratio of static resistance to the dynamic resistance for Ge Diode is 1:400000 & that for Si diode is 1:1000000
2. Explain why the PIV of the semiconductor diode is an important factor.
3. What are the important specifications of a semiconductor diode.
4. What are the difference between signal diode and power diode?

EXPERIMENT NO: 8

TITLE: MEASUREMENT OF POWER IN A SINGLE PHASE AC CIRCUIT USING A WATTMETER.

OBJECTIVES:

To measure the power dissipated in a single phase inductive circuit with the help of a dynamometer type wattmeter and to determine the power factor of the load by measuring the voltage and current in the load.

CIRCUIT DIAGRAM:

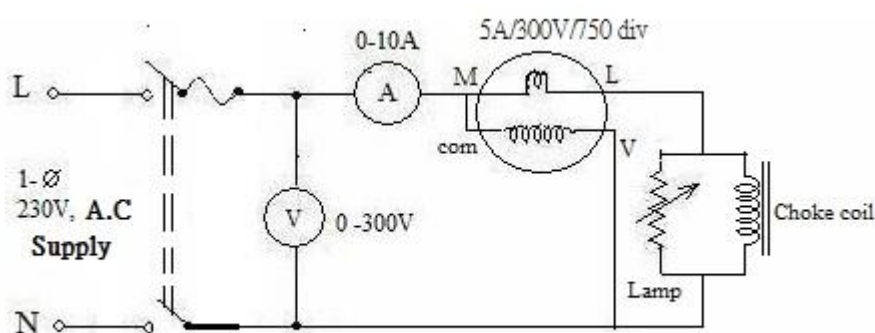


Fig.7

LIST OF APPARATUS:

Serial No	Apparatus	Maker	Type	Range	Quantity

THEORY:

A wattmeter is an indicating instrument used to measure electric power (ac or dc).In the dynamometer type wattmeter there are two sets of coils , current coil (cc) and the pressure coil (pc) . The pressure coil (also known as voltage coil) is the moving coil and is attached to the spindle, which carries the pointer of the instrument. The current coil is connected in series with the load and carries the load current .The pressure coil is connected across the load (or supply) circuit in which power is to be measured.

It can be shown that the mean deflecting torque $T \propto VI \cos\phi$ where V & I are the rms value of voltage & current and ϕ is the angle between V & I (leading or lagging)

Note: The wattmeter is designed to have multiple ratings and hence the scale is not calibrated in watts but in division only .To obtain the power in watts , the readings of the wattmeter are to be multiplied by a factor called the multiplying factor(MF) which is given by

$$MF = \frac{\text{rated voltage} \times \text{rated current}}{\text{No of division on the scale}} \quad (\text{in watt/division})$$

PROCEDURE:

1. Select proper ratings of cc (say 5 or 10 A) and pc (say 125 or 250 V) of the wattmeter.
2. Connect the circuit as shown in the circuit diagram.
3. Take sets of reading by varying the load impedance.
4. Calculate the power factor at each stage.

EXPERIMENTAL OBSERVATION & RESULTS:

Sl No.	Load voltage (V)	Load current (A)	Wattmeter reading		Power factor
			Divisions	Watts(W)	

CONCLUSION:

This is your interpretation of the results of the experiment as an engineer. Be brief and specific. Give reasons for important discrepancies.

REPORTS:

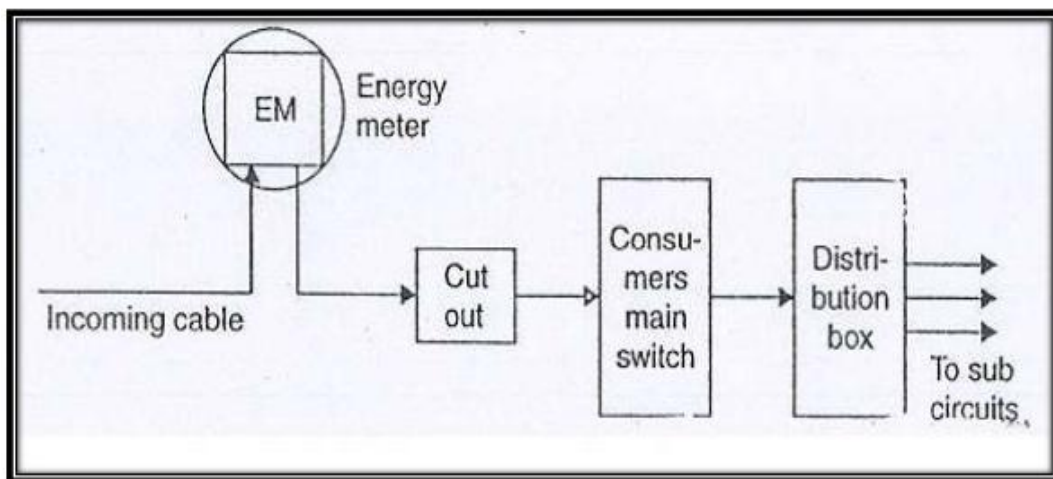
1. Show the relation between p. f. apparent power, true power in an ac circuit and also show the relation between p. f., impedance, reactance and resistance of a circuit.
2. Show that power absorbed by a pure inductor is zero.
What is the p.f. in a purely resistive, inductive and capacitive circuit? State whether the p.f.s are leading or lagging in each case

EXPERIMENT NO. 9

TITLE: DEMONSTRATION OF LAYOUT OF HOUSE WIRING.

DIAGRAM:

Single line diagram of electrical Installation



EXPERIMENT NO. 10

TITLE: DEMONSTRATION OF MEASUREMENT OF INSULATION RESISTANCE .

DIAGRAM :

